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Information Compression Techniques for Control Data Exchange in Joint Communication and Control system

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Abstract

The design of Joint Communication and Control (JCC) systems is critical for modern cyber-physical applications, where strict bandwidth constraints often lead to severe control instability and degraded performance. To address the fundamental challenge of maintaining control accuracy under extreme network limitations, this thesis investigates an intelligent, context-aware information compression strategy utilizing a multi-degree-of-freedom robotic manipulator as a comprehensive testbed. The proposed framework integrates an Event-Triggered Control (ETC) mechanism to reduce communication frequency in the temporal domain with a Deep Reinforcement Learning (DRL) agent–specifically utilizing Proximal Policy Optimization (PPO)– acting as a dynamic bandwidth allocator in the spatial domain. Unlike traditional uniform quantization or static allocation methods, the PPO agent learns a "pragmatic communication" strategy by observing real-time physical states and optimizing a reward function derived from the Linear Quadratic Regulator (LQR) tracking cost. This formulation enables the agent to dynamically distribute a strictly limited bit budget across different robotic joints, proactively sacrificing the precision of low-sensitivity linkages to protect highly sensitive base joints during critical movements. Simulation conducted on a PyBullet-based platform demonstrate the superiority of this co-design, which substantially reduces the overall communication payload–consuming less 6% of the ideal bandwidth–while effectively mitigating trajectory jittering and preserving near-ideal tracking precision, thereby validating the immense potential AI-driven dynamic resource allocation in enhancing the resilience of resource-constraited networked control systems.

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor

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1 Introduction

1.1 Background and Motivation

With the rapid advancement of wireless communication networks and distributed computing architectures, traditional isolated robotic systems are undergoing a profound paradigm shift toward Cloud Robotics and Teleoperation [7]. In these modern applications, the intensive computational and decision-making tasks multi-degree-of-freedom (Multi-DOF) robotic manipulators are increasingly offloaded to edge servers or cloud infrastructure. Such systems, where sensors, controllers, and actuators are interconnected via shared communication networks to form a cyber-physical closed loop, are defined as Networked Control Systems (NCS) [6]. NCS endows robotic systems with unprecedented flexibility, scalability, and collaborative computing capabilities, significantly reducing the hardware costs and local power consumption of individual robots.

However, incorporating communication networks into physical control loops introduces severe engineering and theoretical challenges. To maintain smooth and precise physical trajectory tracking, high-DOF robotic manipulators (e.g., 6-DOF and above) typically require extremely high-frequency state updates, often exceeding 1000 Hz in sensor sampling rates [15]. In stark contrast, real-world wireless communication channels are subject to strict bandwidth constraints and exhibit highly dynamic, unpredictable behaviors. When sensors inject massive volumes of full-precision state data into the network at high frequencies, the limited channel capacity is rapidly exhausted, triggering severe network congestion, packet loss, and unpredictable network-induced delays.

Under extreme network resource constraints, the traditional "Best-Effort" data transmission protocols are fundamentally inadequate for high-precision physical control [5]. Deteriorating communication quality directly disrupts the timeliness and integrity of the control feedback loop, causing severe trajectory jittering, control divergence, and ultimately, system failure during delicate manipulation tasks. Consequently, the critical challenge in modern Cyber-Physical Systems (CPS) lies in drastically reducing the communication data payload under extreme bandwidth limitations while simultaneously guaranteeing the control stability and tracking precision of the robot.

1.2 The Fallacy of Separation and the Need for JCC

Historically, the design of networked control systems has been heavily influenced by Shannons Separation Principle, which posits that source coding (data compression) and channel coding (error protection) can be designed independently without any loss of optimality. In the context of robotic control, this principle translates into a rigid decoupling between communication protocols and control algorithms [14]. Communication engineers typically design systems to minimize the generic reconstruction error of the transmitted dataoften utilizing uniform quantization and periodic transmission mechanismswhile control engineers develop algorithms under the unrealistic assumption of perfect, continuous data reception. However, this artificial separation is fundamentally flawed in resource-constrained cyber-physical systems. Treating all sensory data equally blindly ignores the intrinsic "control-criticality" of the physical states. In multi-DOF robotic manipulators, minor deviations in specific joints can lead to drastically different physical consequences, rendering uniform compression highly inefficient and prone to causing catastrophic system instability [8].

To overcome the limitations of the separation paradigm, this thesis advocates for a Joint Communication and Control (JCC) framework. The core philosophy of JCC is that when com-

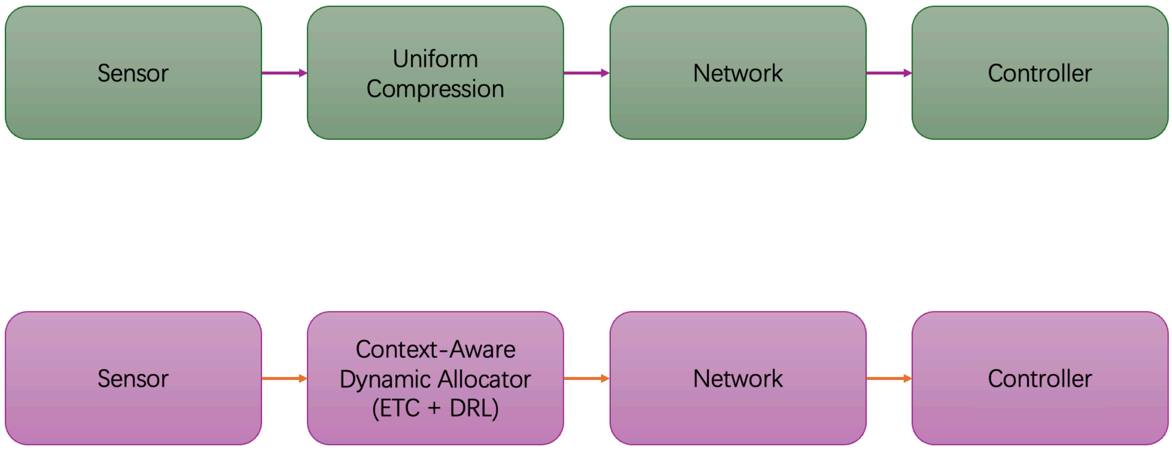

Figure 1: System Paradigm Comparison between Traditional NCS and Proposed JCC.

munication bandwidth is severely restricted, data transmission and compression algorithms must be explicitly control-oriented and context-aware. Instead of blindly minimizing the mathematical mean square error (MSE) of the data stream, the communication protocol must priority the transmission of information that is most vital to maintaining the stability and tracking precision of the physical plant.

Specifically, this thesis implements the JCC paradigm through two complementary technical approaches. In the temporal domain, the rigid, high-frequency periodic transmission is replaced by an Event-Triggered Control (ETC) mechanism [12]. ETC ensures that sensor data is transmitted over the network only when the physical state deviation crosses a predefined safety threshold, effectively eliminating redundant data exchange during stable operational phases. In the spatial domain, the proposed framework exploits the mechanical lever effect inherent in robotic manipulators–where errors in base joints exponentially magnify end-effector deviations compared to errors in distal joints. By dynamically evaluating the real-time sensitivity of each joint, the system executes non-uniform, weighted bandwidth allocation, stripping communication resources from robust linkages to guarantee the high-precision transmission of critical joint states [3].

1.3 AI-Driven Dynamic Allocation

While static, control-aware allocation methodssuch as those based on Linear Quadratic Regulator (LQR) sensitivity matricesoffer significant improvements over uniform compression, they suffer from inherent limitations. Previous research in NCS has extensively explored static resource allocation and Value-of-Information (VoI) based scheduling to optimize bandwidth usage [4]. However, these traditional methodologies largely rely on linearized models and fixed sensitivity weightings. In real-world applications, the dynamics of multi-DOF robotic manipulators are highly nonlinear; the sensitivity of each joint varies dynamically in real-time depending on the robot's current posture, velocity, and external payloads. Consequently, static allocation policies inevitably result in sub-optimal performance during complex, high-speed maneuvers, driving the need for task-oriented or goal-oriented communication frameworks [9].

To overcome the limitations of static models and resolve this complex, nonlinear dynamic resource allocation game, this thesis introduces Deep Reinforcement Learning (DRL) into the communication protocol design. Recently, DRL has demonstrated exceptional capabilities in handling high-dimensional, continuous control and scheduling problems within cyber-physical environments, bridging the gap between optimal control theory and adaptive communication [2]. Specifically, this work employs the Proximal Policy Optimization (PPO) algorithm [10], which is highly favored for its robust convergence properties, sample efficiency, and stability in continuous action spaces.

Driven by the PPO algorithm, the framework realizes the philosophy of "Pragmatic Communication". Instead of following predefined mathematical rules, the AI agent acts as a centralized bandwidth arbiter. It continuously observes the real-time physical tracking errors and environmental states. By optimizing a reward function strictly tied to the minimization of the overall control cost, the agent organically learns a pragmatic strategy. During critical maneuvers, the DRL agent proactively strips communication bandwidth from low-sensitivity linkages and reallocates the strictly limited bit budget to protect highly sensitive base joints. This learning-based dynamic allocation ensures optimal system resilience under extreme network constraints.

1.4 Research Objectives and Contributions

The primary objective of this thesis is to formulate, implement, and evaluate an intelligent, Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework. This framework aims to guarantee precise and stable trajectory tracking for multi-degree-of-freedom (Multi-DOF) robotic manipulators operating over extreme bottleneck communication channelsspecifically under conditions where available bandwidth is severely restricted to less than 10% of the ideal requirement. By seamlessly integrating temporal event-triggered mechanisms with spatial dynamic bandwidth allocation, this research seeks to resolve the fundamental trade-off between communication efficiency and physical control stability in modern networked cyber-physical systems.

To achieve the aforementioned objectives, the core contributions of this thesis are summarized as follows:

1. **Development of a High-Fidelity Cyber-Physical Testbed**: A comprehensive closed-loop control simulation environment was constructed utilizing the PyBullet physics engine. This testbed faithfully replicates realistic kinematic and dynamic constraints of robotic manipulators, providing a robust and reproducible foundation for evaluating networked control algorithms.

2. **Implementation of a Hybrid ETC and LQR-Weighted Benchmark**: A baseline evaluation system was designed by integrating Event-Triggered Control (ETC) in the temporal domain with Linear Quadratic Regulator (LQR)-based static weighted compression in the spatial domain. This benchmark successfully validates the necessity of control-aware, non-uniform data compression over traditional periodic and uniform methods.

3. **Design of a Novel DRL-Driven Context-Aware Allocator**: As the primary innovation, this thesis proposes a dynamic bandwidth allocator powered by the Proximal Policy Optimization (PPO) algorithm. The DRL agent organically learns a "pragmatic communication" strategy by directly observing nonlinear tracking errors and optimizing real-time bandwidth distribution without relying on fixed linear mathematical approximations.

4. **Empirical Validation of Extreme Bandwidth Compression**: Extensive experimental evaluations demonstrate the superiority of the proposed PPO-driven framework. The intelligent agent successfully compressed the communication data payload to under 6%

of the ideal baseline channel capacity, while effectively eliminating trajectory jitter and maintaining near-perfect tracking precision for highly sensitive joints.

1.5 Thesis Organization

2 System Modeling and Methodology

2.1 Overall System Architecture and Cyber-Physical Modeling

The foundation of addressing communication bottlenecks in Networked Control Systems (NCS) lies in accurately modeling the interplay between the physical plant dynamics and digital communication infrastructure.

2.1.1 Networked Control System Architecture

The proposed NCS architecture comprises a closed-loop cyber-physical pipeline encompassing sensors, a bandwidth-limited communication channel, a remote controller, and physical actuators. Unlike traditional tethered robots with dedicated point-to-point analog connections, the sensory and control data in this modern architecture must traverse a shared, bandwidth-limited communication medium.

1. **Sensing and Edge Allocation Node**: High-precision encoders continuously monitor the physical states of the manipulator (e.g., joint angles and angular velocities). Instead of transmitting this full-precision data periodically, an edge computing node actsas the "JCC Allocator". It evaluates the necessity of transmission (Temporal ETC) and dynamically assigns limited quantization bits to different state variables (Spatial Allocation).

2. **Bandwidth-Limited Communication Channel**: The network medium is characterized by strict capacity constraints, modeled as a maximum allowable payload of B_{total} bits per sampling interval. This constraint applies universally, representing physical limits in transmission or artificial rate limits in shared wired networks.

3. **Controller and Zero-Order Hold (ZOH)**: The remote controller receives the heavily quantized or delayed state information. A Zero-Order Hold (ZOH) mechanism is implemented at the receiver side to maintain the control input using the most recently received packet when new data is unavailable (due to ETC silence or network loss).

4. **Actuators**: The calculated control torques are transmitted back to robot's joints motors to execute the desired trajectory.

2.1.2 Robotic Manipulator Dynamics and State-Space Modeling

To derive control-oridented communication strategies, a rigorous mathematical model of the physical plant is required. The continuous-time rigid-body dynamics of an n-degree-of-freedom (DOF) robotic manipulator can be described by the following nonlinear second-order differential equation:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau \tag{1}$$

where:

- $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ represent the vectors of joint angles, angular velocities, and angular accelerations, respectively.
- $M(q) \in \mathbb{R}^{n\times n}$ is the symmetric, positive-definite inertial matrix.
- $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ represents the Coriolis and centrifugal force matrix.
- $G(q) \in \mathbb{R}^n$ is the gravity loading vector.

- $\tau \in \mathbb{R}^n$ is the vector of control torques applied by the actuators.

To transform this into a standard state-space representation, let the system state vector be defined as $x(t) = [q(t)^T, \dot{q}(t)^T]^T \in \mathbb{R}^{2n}$, and the control input as $u(t) = \tau(t) \in \mathbb{R}^n$. The nonlinear dynamics can be rewritten as $u(t) = \tau(t) \in \mathbb{R}^n$. The nonlinear dynamics can be rewritten as a first-order ordinary differential equation $\dot{x}(t) = f(x(t), u(t))$:

$$\dot{x}(t) = \frac{d}{dt}\begin{bmatrix} q \\ \dot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ M(q)^{-1}(\tau - C(q,\dot{q})\dot{q} - G(q)) \end{bmatrix} \tag{2}$$

For advanced linear control methodologies (such as LQR) and sensitivity analysis, this nonlinear model is linearized around a nominal operating trajectory or equilibrium point (x_e, u_e). Using Jacobian linearization, the continuous-time linear time-invariant (LTI) model is obtained:

$$\dot{\tilde{x}}(t) = A_c\tilde{x}(t) + B_c\tilde{u}(t) \tag{3}$$

where $x_k \in \mathbb{R}^{2n}$ is the discrete state at time step k,$u_k \in \mathbb{R}^n$ is the discrete control input, and $w_k \in \mathbb{R}^{2n}$ represents the additive process noise (e.g., modeling inaccuracies and physical disturbances), assumed to be zero-mean Gaussian noise with covariance W. The discrete matrices are $A = e^{A_c T_s}$ and $B = \int_0^{T_s} e^{A_c t} B_c dt$. Physically, matrix A captures the system's inertia and internal state couping, while matrix B maps the applied actuator torques to the resulting state transitions.

2.1.3 Bandwidth-Limited Communication Network Model

The critical constraint in this thesis is the extreme limitation of the communication channel. Let the network capacity limit be strictly defined as a maximum allowed budget of B_{total} bits per sampling interval T_s.

At any discrete time step k, if the sensor decides to transmit data, it can not send the ideal 64-bit floating-point representation of x_k. Instead, the state vector must be quantized. Let $b_(i,k)$ denote the number of bits allocated to the i-th element of the state vector x_k. The resource constraint mandates that:

$$\sum_{i=1}^{2n} b_{i,k} \leq B_{total} \tag{4}$$

Let $\mathcal{Q}(\cdot)$ denote the quantization function. The transmitted and subsequently received quantized state is denoted as $x_k^q = \mathcal{Q}(x_k, b_k)$, introducing a quantization error $e_(q,k) = x_k - x_k^q$. If $b_(i,k) = 0$, the i-th state is entirely discarded during transmission.

Furthermore, to model the temporal aspect the transmission, an indicator variable $\gamma_k \in \{0,1\}$ is introduced, where $\gamma_k = 1$ indicates a packet is triggered and successfully transmitted, and $\gamma_k = 0$ indicates no transmission (event-triggered silence).

At the controller side, a Zero-Order Hold (ZOH) updates the estimated state $\hat{x}_k$ based on the communication status:

$$\hat{x}_k = \begin{cases} x_k^q, & \text{if } \gamma_k = 1 \text{ (Packet received)} \\ \hat{x}_{k-1}, & \text{if } \gamma_k = 0 \text{ (No transmission, hold previous state)} \end{cases} \tag{5}$$

This can be concisely written as the receiver update equation:

$$\hat{x}_k = \gamma_k x_k^q + (1-\gamma_k)\hat{x}_{k-1} \tag{6}$$

The control input u_k is then strictly computed based on this remote estimate $\hat{x}_k$ rather the true physical state x_k. The central challenge addressed in the following sections is how to simultaneously optimze γ_k (when to transmit) and b_k (how to allocate bits) to minimize the deviation between the actual physical trajectory and the desired reference.

2.2 Temporal Domain: Event-Triggered Control Mechanism

In Networked Control Systems, the temporal strategy of when to transmit data is as critical as the spatial strategy of how to compress it. This section details the deployment of an Event-Triggered Control (ETC) mechanism designed to aggressively truncate redundant transmissions in the temporal domain.

2.2.1 The Inefficiency of Traditional Time-Triggered Sampling

Traditionally, digital control systems rely on Time-Triggered Control (TTC), where sensor measurements are sampled and transmitted at rigidly fixed, periodic intervals $t_k = kT_s$. While TTC simplifies system design and analysis, it is notoriously inefficient regarding bandwidth utilization.

As rigorously demonstrated by Åström and Bernhardsson [1], Lebesgue sampling (event-based) significantly outperforms Riemann sampling (time-based) in stochastic control systems. Under a TTC paradigm, the sensor continuously broadcasts data even when the robotic manipulator has reached a stable equilibrium or is executing a steady-state motion with negligible tracking error. In a strictly bandwidth-constrained shared network, shared network, this continuous stream of redundant, non-informative data squanders the limited channel capacity, directly exacerbating network congestion and packet loss for other critical nodes. To achieve "pragmatic communication," the system must abandon periodic transmission and adopt a paradigm of "speaking only when necessary".

2.2.2 Design and Derivation of the Event-Triggering Condition

The fundamental logic of Event-Triggered Control is to continuously evaluate the "novelty" or "urgency" of the current physical state at the sensor side and initiate a transmission only when the state deviation threatens the system's stability.

Let $\hat{x}_k$ represent the estimated state maintained at the remote controller via the Zero-Order Hold (ZOH). The measurement error (or event error), denoted as e_k, is defined as the discrepancy between the actual current state x_k at the sensor and the remote estimate $\hat{x}_k$:

$$e_k = x_k - \hat{x}_k \tag{7}$$

The triggering condition is governed by a threshold mechanism. A transmission is triggered if and only if the norm of the measurement error exceeds a predefined safety threshold δ:

$$||e_k||_2 > \delta \tag{8}$$

Mathematical Derivation of Stability: The viability of this threshold is not arbitrary; it is strictly bounded by Lyapunov stability theory. Based on the ETC framework established by Tabuada [13], we must ensure that the control system maintains Asymptotic Stability despite the discontinuous state updates.

Assume there exits a stabliziing state-feedback controller $u_k = -K\hat{x}_k$, and a corresponding discrete-time Control Lyapunov Function $V(x_k) = x_k^T P_L x_k$ (where P_L is a symmetric positive-define matrix). For the closed-loop system to be asymptotically stable, the Lyapunov difference must be strictly negative define:

$$\Delta V = V(x_{k+1}) - V(x_k) < 0 \tag{9}$$

By substituting the closed-loop dynamics $x_{k+1} = Ax_k - BK\hat{x}_k = (A - BK)x_k - BKe_k$ into the Lyapunov difference, ΔV can be bounded by a function of the true state x_k and the measurement error e_k. According to Input-to-State Stability (ISS) properties, ΔV takes the form:

$$\Delta V \leq -\alpha(||x_k||) + \beta(||e_k||) \tag{10}$$

where $\alpha(\cdot)$ and $\beta(\cdot)$ are class $\mathscr{K}$ functions. To guarantee $\Delta V < 0$, the error must satisfy $\beta(||e_k||) < \alpha(||x_k||)$. This implies that as long as the measurement error e_k is bounded by a relative threshold (e.g., $||e_k|| \leq \sigma||x_k||$) or a sufficiently small absolute static threshold δ, the stability of the robotic manipulator is mathematically guaranteed.

In our proposed framework, to simplify the edge computation load on the sensor node, a robust absolute static threshold $||e_k||_2 > \delta$ is implemented as the primary transmission trigger.

2.2.3 Mathematical Abstraction of Communication Behavior

To formally integrate the temporal ETC behavior with the spatial quantization algorithm (discussed in Section 3.3), we introduce a Boolean communication decision variable $\gamma_k \in \{0, 1\}$.

The edge sensor node continuously evaluates the ETC condition and generates the decision variable:

$$\gamma_k = \begin{cases} 1, & \text{if } ||x_k - \hat{x}_{k-1}||_2 > \delta \quad \text{(Trigger Event)} \\ 0, & \text{otherwise} \quad \text{(Silence)} \end{cases} \tag{11}$$

Consequently, the mathematical model of the receiver (the Zero-Order Hold at the controller) is perfectly unified. Upon integrating the quantization process $x_k^q = \mathscr{Q}(x_k, b_k)$, the remote state estimate $\hat{x}_k$ is updated according to the following equation.

$$\hat{x}_k = \gamma_k x_k^q + (1 - \gamma_k)\hat{x}_{k-1} \tag{12}$$

This equation powerfully summarizes the JCC temporal-spatial interplay: If $\gamma_k = 1$, the network is actively utilized, and the spatial quantizer $\mathscr{Q}(\cdot)$ determines the precision of the payload x_k^q. If $\gamma_k = 0$, exactly zero bits are transmitted over the channel, and the controller relies purely on the historical memory $\hat{x}_{k-1}$, thereby achieving massive bandwidth savings in the temporal domain.

2.2.4 Spatial Domain: Static LQR-Weighted Data Compression

While the temporal ETC mechanism dictates when to transmit, the spatial compression mechanism dictates *how* to allocate the severely limited bit budget B_{total} across multiple joint states during an active transmission. Traditional approaches uniformly divide the bit budget ($b_i = B_{total}/n$), which blindly treats a 0.1-rad error at the base joint as equivalent to a 0.1-rad error at the end-effector.

To overcome this, a static control-aware baseline utilizes the infinite-horizon Linear Quadratic Regulator (LQR) cost function $J = \sum(x_k^T Q x_k + u_k^T R u_k)$. By solving the Discrete Algebraic Riccati Equation (DARE), the system extracts the Hessian sensitivity matrix P. According to rate-distortion theory, the optimal bit allocation b_i that minimizes the control-oriented distortion is formulated as:

$$b_i = \frac{B_{total}}{n} + \frac{1}{2}\log_2\left(\frac{P_{ii}\sigma_i^2}{\left(\prod_{j=1}^{n} P_{jj}\sigma_j^2\right)^{1/n}}\right) \tag{13}$$

This static LQR-weighted allocation significantly improves control resilience by permanently stripping bits from low-sensitivity joints and granting them to high-sensitivity joints (e.g., base links). However, in highly nonlinear robotic dynamics, the true sensitivity of a joint is state-dependent and fluctuates drastically based on the manipulator's real-time posture and velocity. To achieve true context-awareness, the system must transition from static mathematical allocation to dynamic, learning-based arbitration.

2.3 Deep Reinforcement Learning for Pragmatic Communication

The core innovation of this thesis is the formulation of a dynamic, context-aware bandwidth allocator driven by Deep Reinforcement Learning (DRL). Rather than relying on fixed linear approximations (such as the static P matrix), an AI agent is deployed at the edge sensor node to act as a centralized communication arbiter. The agent observes the real-time physical tracking errors and dynamically redistributes the strictly limited B_{total} bits at every time step. This learning-based paradigm epitomizes "Pragmatic Communication"the communication protocol evolves explicitly to serve the downstream physical task, learning to proactively sacrifice the precision of robust linkages to guarantee the survival of critical joints.

2.3.1 Markov Decision Process (MDP) Formulation

To enable DRL training, the dynamic bandwidth allocation problem must be strictly formulated as a Markov Decision Process (MDP), defined by the tuple $\langle \mathscr{S}, \mathscr{A}, \mathscr{P}, \mathscr{R}, \gamma \rangle$.

1. State Space (Observation) $\mathscr{S}$: For the agent to make context-aware decisions, the observation space must encompass sufficient information regarding the current physical urgency of the robotic manipulator. Relying solely on the raw state x_t is insufficient, as the agent would remain ignorant of the control objective. Therefore, the observation $s_t \in \mathscr{S}$ is constructed as an 8-dimensional continuous vector containing the real-time physical states, the target reference states, and the current tracking errors:

$$s_t = \left[q_t^T,\ \dot{q}_t^T,\ q_{target,t}^T,\ (q_t - q_{target,t})^T \right]^T \in \mathbb{R}^8 \tag{14}$$

This augmented state space allows the neural network to implicitly infer both the current nonlinear posture of the manipulator and the instantaneous tracking deviation, which fundamentally dictate the real-time sensitivity of each joint.

2. Action Space $\mathscr{A}$ **and Discrete Bit Mapping**: The primary objective of the agent is to allocate the integer bit budget B_{total} across the n state variables. Directly outputting discrete integers via reinforcement learning introduces non-differentiable operations and combinatorial explosion, severely hindering convergence. To resolve this, a continuous action space is adopted, combined with a deterministic Softmax mapping layer.

The DRL agent outputs a continuous action vector $a_t = [a_{1,t}, a_{2,t}, \dots, a_{n,t}]^T \in [-1, 1]^n$. To map these continuous unbounded preferences into a strict constraint $sum_{i=1}^{n} b_i = B_{total}$ where $b_i \in \mathbb{Z}^+$, the following transformation is applied:

First, the actions are converted into a probability distribution p_i using Softmax function:

$$p_i = \frac{\exp(a_{i,t})}{\sum_{j=1}^{n} \exp(a_{j,t})} \tag{15}$$

Next, the probability distribution is scaled by the total bit budget to obtain the floating-point bit allocation:

$$b_{float,i} = p_i \times B_{total} \tag{16}$$

Finally, the floating-point values are rounded to the nearest integer $b_i = [b_{float}, i]$. To strictly enforce the resource constraint due to rounding sequentially assigned to the state variables exhibiting the largest fractional remainder $(b_{float,i} - b_i)$; conversely, if the sum is excessive, bits are iteratively deducted from the states holding the maximum allocation. This mapping guarantees that the strict network constraint is never violated while maintaining a smooth gradient landscape for DRL training.

3. Reward Function Design $\mathscr{R}$: The reward function is the guiding signal that shapes the "pragmatic" intelligence of the agent. Since the bandwidth constraint B_{total} is strictly enforced by the action mapping layer, there is no need to penalize the agent for bandwidth consumption during spatial allocation. Instead, the reward r_t is purely and directly tied to the downstream physical control performance.

At each time step, the instantaneous reward is defined as the negative weighted tracking error (incorporating the LQR Q matrix):

$$r_t = -\left((x_t - x_{target,t})^T Q(x_t - x_{target,t})\right) \tag{17}$$

For a 2-DOF manipulator where base joint is exponentially more critical than the shoulder joint, the Q matrix is highly skewed (e.g., $Q = \mathrm{diag}(1000, 1)$). Because the DRL seeks to maximize the cumulative expected reward $\mathbb{E}[\sum \gamma^t r_t]$, it quickly discovers a counter-intuitive but mathematically optimal strategy: it will willingly output actions that allocate 0 bits to the shoulder joint (allowing its local error to accumulate), in order to secure maximum bits for the base joint to prevent the massive–1000 penalty multiplier. This mechanism is the mathematical engine behind "Pragmatic Communication".

2.3.2 PPO Algorithm and Policy Network Architecture

Given the continuous nature of the action space and requirement for stable monotonic convergence, the Proximal Policy Optimization (PPO) algorithm [11] is selected over value-based methods (like DQN) or traditional Policy Gradients (like DDPG). PPO operates under an Actor-Critic architecture, utilizing two separate neural networks:

1. **The Actor Network (Policy Function $\pi_\theta(a_t|s_t)$)**: Parameterized by weights θ, this network maps the current state s_t to Gaussian probability distribution over the continuous actions a_t, from which the agent samples its allocation decisions.

2. **The Critic Network (Value Function $V_\phi(s_t)$)**: Parameterized by weights ϕ, this network estimates the expected cumulative discounted reward from state s_t, serving as baseline to evaluate the quality of the Actor's actions.

Both the Actor and Critic networks are constructed using Multilayer Perceptrons (MLPs). The input layer ingests the 8-dimensional state vector s_t. This is followed by two hidden layers, each comprising 128 neurons activated by the ReLU (Rectified Linear Unit) non-linearity, ensuring sufficient representational capacity to capture the highly nonlinear relationship between robotic kinematics and bandwidth sensitivity. To update the Actor network, PPO defines a clipped surrogate objective function. Left the advantage function $\hat{A}_t = R_t - V_\phi(s_t)$ represent how much better an action was compared to Critic's expectation. Let $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ denote the probability ratio between the updated policy and the old policy. The core objective maximized by PPO is:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\right)\right] \tag{18}$$

where ε is a hyperparameter (typically 0.2). The clip function restricts the policy update step size. If a new allocation strategy yields exceptionally high rewards, PPO will not greedily shift the neural network weights too drastically, preventing the policy from collapsing into suboptimal deterministic extremes (e.g., permanently assigning all bits to one joint regardless of the state). This trust-region optimization is paramount for training stability in cyber-physical environments, where extreme exploration can lead to immediate system divergence.

2.3.3 Training Stability and Hard Negative Mining Strategy

Training a DRL agent in a closed-loop control system presents severe challenges regarding state exploration. If the initial DRL agent randomly allocates bits uniformly, the LQR controller might successfully stabilize the robot near the target trajectory. In this safe, steady-state region, the errors are negligible, and the rewards r_t approach zero. The agent suffers from a "vanishing signal problem"– it learns that *any* bit allocation works fine when robot is safe, and thus fails to learn the critical "survival prioritization" required during extreme physical disturbances.

To overcome this, a **Hard Negative Mining (Edge State Initialization)** strategy is explicitly implemented within the training pipeline. During the `reset()` phase of each training episode, the robotic manipulator is not initialized at the safe origin or along the reference trajectory. Instead, the joints are deliberately spawned at extreme adversarial positions (e.g., q_1, q_2 injected with maximum permissible initial deviations).

This adversarial initialization forces the environment to immediately yield catastrophic physical costs (massive negative rewards) at $t = 0$. To survive the episode without triggering the safety truncation threshold (which aborts the episode and administers an additional–500 penalty), the PPO agent is forced to rapidly discover the causal link between base-joint tracking errors and system collapse. It effectively "learns under duress", heavily accelerating the convergence towards the optimal pragmatic communication strategy.

3 Discussion

The discussion is your opportunity to impress an expert with your depth of understanding. Don't worry so much about the non-expert reader – she or he can skip to the Conclusions.

For a research project this section should provide a logical argument leading from the experimental observations to the final conclusions. Evaluate your results thoroughly and gain every possible piece of understanding from them. Compare the results in detail to other work in the field. This should be a *critical* comparison so don't just say that your results are different from previous work; explain *why*. Never say 'The results were as expected' unless you have already described exactly what was expected!

The main purpose of the Discussion in a design-and-build project is to assess the performance of your product against the specification, showing its strengths and weaknesses.

4 Conclusions

Every report must have conclusions, built on the discussion. This section includes a summary of the main achievements of the work but is more than that. The noun ‘conclusion’ can be defined as ‘a judgement or decision reached by reasoning’ [**oupconc**] so you should highlight what has been learnt as a result of your project. What is the big picture that the reader should take away?

The Conclusions should include an assessment of the outcome against the original aims of the project. If you were unable to fulfil some aims, explain why. It is also useful to review the project plan (which should be included with the report). Did some tasks prove to be unexpectedly difficult, for instance?

4.1 Suggestions for further work

Almost every project leaves you with ideas for the future. Research typically answers some questions while opening new ones; by the end of a design-and-build project you may have thought of a superior approach. Examiners are impressed by intelligent suggestions for further work because they show that you really understand the project. It is often a sign of strength to identify the weak points and suggest solutions for them.

References

[1] Marun K. Mitra, *Effective Technical Communication: A Guide for Scientists and Engineers*, Oxford (2006).

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[3] Trevor M. Young, *Technical Writing A–Z: A Commonsense Guide to Engineering Reports and Theses*, American Society of Mechanical Engineers (2005).

[4] G M Fitzmaurice and D J Hand, *A comparison of two average conditional error estimators*, Pattern Recognition Letters, **6**, 221–228 (1987).

[5] Freescale Semiconductor, *Datasheet for MC9S08QG microcontroller*, revision 4 (2008).

[6] Morgan Advanced Ceramics, *Advanced Ceramics: Silicon Nitride*, `www.morgantechnicalceramics.com` (viewed 2009 May 13).

[7] Oxford University Press, *AskOxford*, `www.askoxford.com/?view=uk` (viewed 2009 May 13).

‘Conclusion’ is defined as (1) an end or finish, (2) the summing-up of an argument or text, (3) a judgement or decision reached by reasoning or (4) the settling of a treaty or agreement.

A Appendices

Use appendices for supporting material which is relevant but of a detailed nature. In this way the flow of ideas in the body of the report is not interrupted. Appendices are usually lettered rather than numbered to distinguish them. Several uses for appendices have been suggested already but bear in mind that a CD may be more appropriate for large tables of results, long programs and data sheets or other background information. Appendices are not included in the word count because the assessors are not obliged to read them. Do not put vital points here!

Include your project plan and interim reports as appendices on paper.

B University's Plagiarism Statement

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